

Reducing Hallucinations in Large Language Models via Retrieval-Augmented Generation: A Controlled Study on Small-Scale Corpora

Nabarun Kar
Texas A&M University
nabarunkar@tamu.edu

Abstract

Large Language Models (LLMs) are prone to hallucination, generating confident but factually incorrect responses. Retrieval-Augmented Generation (RAG) has emerged as a promising approach to mitigate this issue by grounding model outputs in external knowledge sources. While prior work has focused on large-scale corpora, the behavior of RAG in small, controlled datasets remains underexplored.

In this work, we present a systematic study of RAG performance on a curated corpus of 100 documents augmented with synthetic distractors. We construct a reproducible question-answer benchmark and evaluate multiple retrieval configurations, including dense (FAISS), sparse (BM25, TF-IDF), and hybrid methods, alongside variations in embedding models and chunking strategies. Compared to a no-retrieval baseline (18.4% hallucination, 81.6% accuracy), the best-performing RAG configuration reduces hallucination to 6.6% and achieves 93.4% accuracy, representing an absolute improvement of 11.8 percentage points.

Our results show that retrieval quality is the primary determinant of RAG effectiveness. Dense retrieval (FAISS) achieves the highest accuracy (90.7%), while hybrid retrieval offers greater robustness to distractors, and BM25 provides the highest recall. We further identify a non-linear relationship between chunk size and performance, with intermediate chunk sizes (512 tokens) yielding optimal results. Notably, we uncover a fundamental tradeoff between accuracy and robustness: the best-performing embedding model (E5-Small, 93.4%) is also the most susceptible to distractor contamination (82%), whereas more robust models exhibit lower peak accuracy.

These findings demonstrate that while RAG substantially reduces hallucination in small-scale settings, its effectiveness depends critically on retrieval design choices. This work provides a controlled, fine-grained analysis of RAG behavior and offers practical guidance for building reliable retrieval-augmented systems under resource constraints.

1 Introduction

Large Language Models (LLMs) have demonstrated remarkable capabilities across a wide range of natural language tasks. However, they remain prone to hallucination, producing fluent but factually incorrect or unverifiable outputs. This limitation poses a significant challenge for applications requiring high factual reliability, such as education, scientific analysis, and domain-specific information retrieval.

Retrieval-Augmented Generation (RAG) has emerged as a prominent approach to mitigate hallucination by grounding model outputs in external knowledge sources. In a typical RAG pipeline, a retriever selects relevant documents from a corpus, which are then provided as context to the language model during answer generation. Prior work has shown that RAG can improve factual accuracy, particularly in large-scale settings with extensive knowledge bases.

Despite these advances, existing research predominantly focuses on large corpora and high-resource environments. In contrast, many real-world deployments involve small, domain-specific datasets, where retrieval errors are amplified and system behavior may differ significantly. In such settings, factors such as chunking strategy, embedding model selection, and the presence of misleading or conflicting information can have a disproportionate impact on performance. However, these effects remain insufficiently studied in a controlled and systematic manner.

In this work, we present a comprehensive study of RAG behavior in small, controlled datasets. We construct a curated corpus of 100 documents, including both factual sources and adversarial distractors, and develop a benchmark of 100 question-answer pairs designed to probe different failure modes. Using this framework, we systematically

evaluate the impact of retrieval methods, embedding models, and chunking strategies on hallucination rates and answer accuracy.

Our findings demonstrate that while RAG substantially reduces hallucination, its effectiveness depends critically on retrieval configuration. We identify a non-linear relationship between chunk size and performance, and uncover a fundamental tradeoff between accuracy and robustness to distractors in embedding models. Notably, state-of-the-art embeddings do not consistently outperform simpler baselines in small-scale settings, challenging assumptions derived from large-scale benchmarks.

The contributions of this work are as follows:

- We introduce a controlled experimental framework for evaluating RAG systems on small-scale datasets with adversarial distractors.
- We provide a systematic analysis of retrieval methods, embedding models, and chunking strategies, quantifying their impact on hallucination and accuracy.
- We identify key failure modes, including hallucination despite correct context and distractor-induced errors.
- We uncover a fundamental accuracy–robustness tradeoff in embedding selection for small-corpus RAG systems.
- We offer practical guidelines for designing reliable RAG pipelines under resource and data constraints.

Overall, this work bridges the gap between large-scale RAG research and practical small-domain deployments, providing both empirical insights and actionable recommendations for improving LLM reliability.

2 Methodology

2.1 Dataset Construction

To evaluate Retrieval-Augmented Generation (RAG) under controlled conditions, we constructed a custom corpus consisting of 200 documents. This corpus includes 100 original Wikipedia articles spanning domains such as science, history, and geography, and 100 synthetic distractor documents.

The distractor documents were designed as adversarial noise, containing plausible but incorrect information such as altered dates, swapped entities, and misleading quantitative values. This setup enables systematic evaluation of model robustness when exposed to conflicting or misleading context, rather than purely irrelevant noise.

2.2 Question–Answer Benchmark

A benchmark of 100 human-verified question–answer pairs was developed to evaluate model performance. The questions were categorized into three groups:

- **Conflict (50%)**: Questions where distractor documents contain incorrect but plausible answers that directly contradict the ground truth.
- **Common (30%)**: Standard factual questions where the correct answer is clearly present in the corpus.
- **Missing (20%)**: Questions where the correct answer is not present in the retrieved context, designed to test hallucination tendencies.

This distribution emphasizes adversarial evaluation while maintaining coverage of standard and edge-case scenarios.

2.3 RAG Pipeline Implementation

The RAG pipeline was implemented using LangChain and FAISS. Documents were split into chunks using a recursive character-based splitter, and embeddings were generated using various sentence transformer models. Retrieved documents were passed as context to a large language model (LLM), which generated answers constrained to the provided context.

For all experiments, a fixed retrieval parameter of $\text{top-}k = 5$ was used to ensure consistency across configurations. An overview of the complete system architecture is shown in Figure 1.

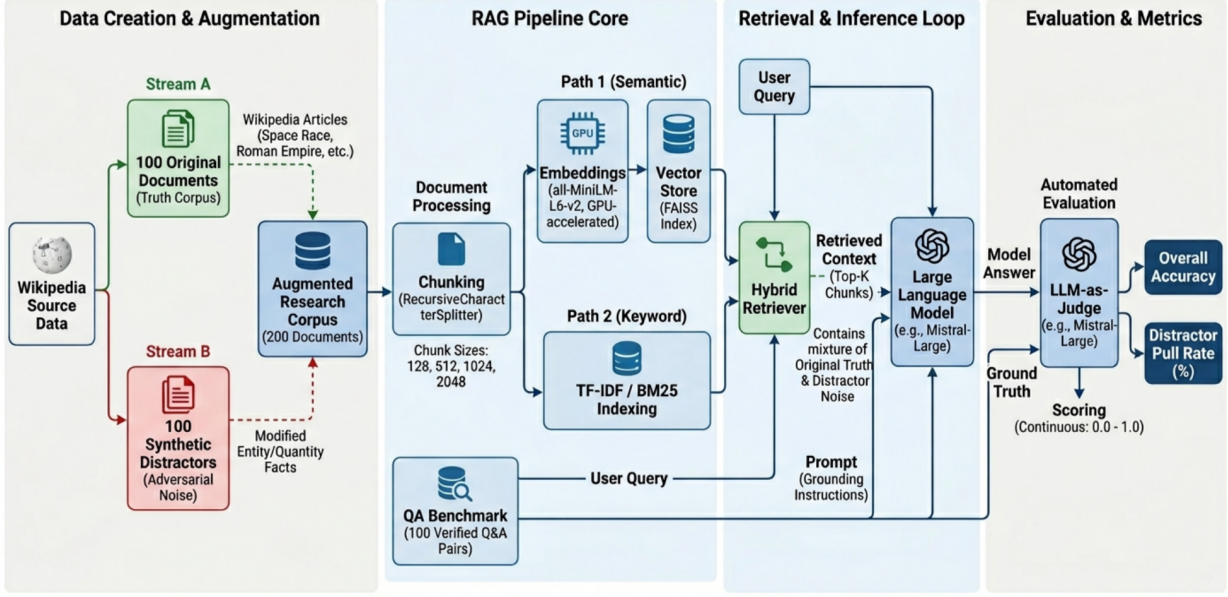


Figure 1: System architecture of the proposed RAG pipeline. The framework consists of (1) controlled data construction with synthetic distractors, (2) dual-path retrieval using semantic (FAISS) and keyword-based (TF-IDF/BM25) methods, (3) retrieval-augmented generation with top-k context, and (4) automated evaluation using an LLM-as-judge.

2.4 Experimental Design

We conducted a series of controlled experiments by varying one component at a time while keeping all others constant. The primary variables investigated include:

- **Embedding Models:** MiniLM-L6 (baseline), BGE-Small, MPNet-Base, and E5-Small.
- **Chunking Strategies:** Chunk sizes of 128, 256, 512, 1024, and 2048 tokens, with proportional overlap.
- **Retrieval Methods:** Dense retrieval (FAISS), sparse retrieval (BM25, TF-IDF), and hybrid combinations.

MiniLM-L6-v2 was selected as the baseline embedding model due to its widespread adoption, lightweight footprint, and strong performance-to-cost ratio. A default chunk size of 512 tokens was used for most experiments, as it represents a balance between providing sufficient contextual information and minimizing retrieval noise.

2.5 Evaluation Framework

Model outputs were evaluated using an automated LLM-as-a-judge framework. For each question, the generated answer was compared against a ground truth reference using a secondary LLM, which assigned a continuous score in the range $[0, 1]$.

A score of 1.0 indicates a fully correct answer, while 0.0 indicates a completely incorrect or hallucinated response. Intermediate values capture partial correctness.

In addition to answer accuracy, we track whether retrieved contexts contain distractor documents, enabling analysis of retrieval robustness and susceptibility to adversarial noise.

2.6 Implementation Details

All experiments were conducted in a GPU-accelerated environment. The primary inference model used across experiments was Mistral-Large, with Mistral-Small used for limited comparative analysis to evaluate performance consistency across model scales.

To ensure reproducibility and stability, evaluation pipelines incorporated checkpointing and retry mechanisms for API calls. Each experimental configuration was executed independently, and results were stored for subsequent analysis. The complete implementation, dataset construction pipeline, and experimental framework are publicly available at: <https://github.com/NabarunKar/rag-hallucination-benchmark>.

3 Evaluation Metrics

To assess the effectiveness of Retrieval-Augmented Generation (RAG) in reducing hallucination, we employ a combination of quantitative metrics that capture both answer correctness and retrieval behavior.

3.1 Answer Accuracy

Answer accuracy serves as the primary performance metric. For each question, the generated response is evaluated against a ground truth reference using an automated LLM-as-a-judge framework, which assigns a continuous score in the range $[0, 1]$.

A score of 1.0 indicates a fully correct answer, while 0.0 indicates a completely incorrect response. Intermediate values capture partial correctness. Overall accuracy is computed as the mean score across all evaluation samples.

3.2 Hallucination Rate

We define hallucination rate as the proportion of incorrect or unsupported responses generated by the model. Since accuracy is computed as a continuous score, hallucination rate is derived as:

$$\text{Hallucination Rate} = 1 - \text{Accuracy} \quad (1)$$

This formulation directly captures the frequency of factual errors and aligns with the primary objective of reducing hallucinations in LLM outputs.

3.3 Distractor Pull Rate

To measure retrieval robustness, we introduce *Distractor Pull Rate*, defined as the proportion of queries for which at least one retrieved document originates from the distractor set.

Formally, for each query, if any of the top- k retrieved documents is labeled as a distractor, the query is counted as a distractor pull. The overall rate is computed as the fraction of such queries across the evaluation set.

This metric provides insight into the susceptibility of retrieval systems to adversarial noise and allows us to analyze the relationship between retrieval quality and downstream model performance.

3.4 Retrieval Recall

To validate retrieval effectiveness, we report Recall@5 on a subset of experiments. Recall@5 measures whether the correct source document is present within the top- $k = 5$ retrieved results.

While not the primary focus of this study, Recall@5 provides a sanity check for retrieval coverage and helps contextualize performance differences across retrieval methods.

3.5 Metric Summary

Together, these metrics provide a comprehensive evaluation framework: accuracy and hallucination rate capture answer correctness, while distractor pull rate and recall characterize retrieval behavior. This combined approach enables a fine-grained analysis of how retrieval quality influences hallucination in small-scale RAG systems.

4 Results

We present the results of our controlled experiments evaluating the impact of retrieval, embedding models, and chunking strategies on hallucination reduction in small-scale RAG systems.

4.1 Overall Performance

Table 1 summarizes the performance across all major configurations. Compared to the no-retrieval baseline (81.6% accuracy), all RAG-based approaches significantly improve answer quality.

The best-performing configuration, as shown in Figure 2, using E5-Small embeddings with FAISS retrieval, achieves 93.4% accuracy, representing an absolute improvement of 11.8 percentage points over the baseline. This corresponds to a reduction in hallucination rate from 18.4% to 6.6%.

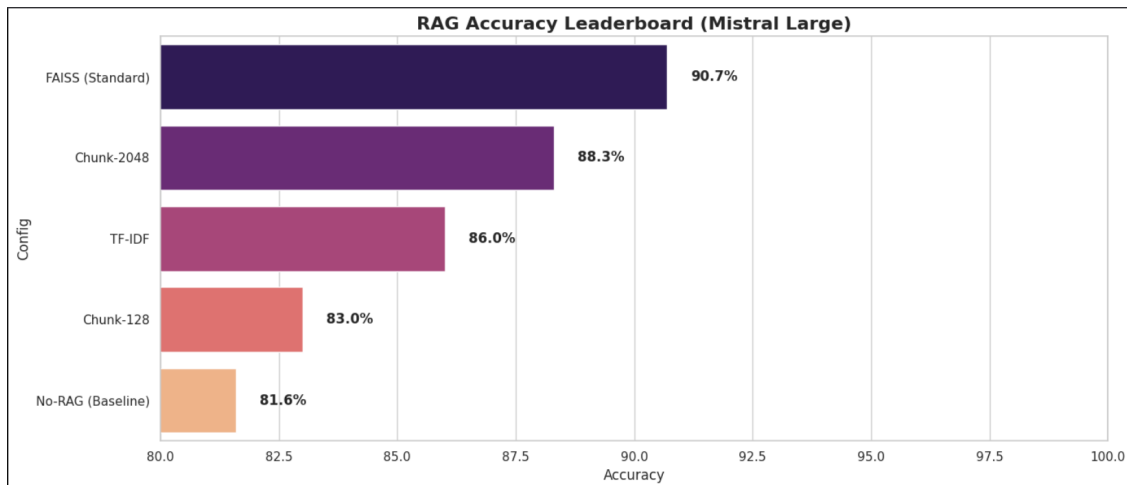


Figure 2: Accuracy comparison across key RAG configurations. Retrieval-augmented methods significantly outperform the no-retrieval baseline, with FAISS-based configurations achieving the highest accuracy.

Configuration	Accuracy (%)	Distractor Pull (%)
E5-Small (FAISS)	93.4	82.0
FAISS (Large, 512)	90.7	78.0
MiniLM-L6 (FAISS)	90.5	78.0
Hybrid (FAISS+TF-IDF)	90.0	71.0
BGE-Small (FAISS)	89.5	78.0
Chunk-1024	89.5	—
Overlap-0%	88.8	—
Chunk-2048	88.3	73.0
BM25-only	87.7	—
Hybrid (BM25+FAISS)	87.7	—
TF-IDF (Large)	86.0	79.0
MPNet-Base (FAISS)	86.0	73.0
Chunk-256	84.5	—
FAISS (Small)	84.0	78.0
Chunk-128	83.0	78.0
No-RAG (Large)	81.6	0.0
No-RAG (Small)	78.0	0.0

Table 1: Performance comparison across retrieval methods, embeddings, and chunking configurations.

4.2 Impact of Retrieval Methods

Retrieval strategy plays a critical role in overall performance. Dense retrieval using FAISS achieves the highest accuracy (90.7%), outperforming both sparse (TF-IDF: 86.0%) and BM25-based approaches (87.7%).

Hybrid retrieval methods offer a strong tradeoff between performance and robustness. The FAISS+TF-IDF hybrid achieves 90.0% accuracy while reducing distractor pull rate to 71.0%, the lowest among high-performing configurations. BM25 achieves near-perfect Recall@5 but lower end-to-end accuracy, indicating that retrieval coverage alone is insufficient for optimal performance.

4.3 Effect of Embedding Models

Embedding model choice significantly influences RAG performance, as illustrated in Figure 3. As shown in Table 2, E5-Small achieves the highest accuracy (93.4%), outperforming all other models.

Embedding Model	Accuracy (%)	Distractor Pull (%)
E5-Small	93.4	82.0
MiniLM-L6	90.5	78.0
BGE-Small	89.5	78.0
MPNet-Base	86.0	73.0

Table 2: Comparison of embedding models under identical experimental conditions.

Interestingly, the state-of-the-art BGE-Small model underperforms the simpler MiniLM baseline. Furthermore, higher-performing embeddings exhibit increased susceptibility to distractor retrieval. E5-Small, while achieving the highest accuracy, also has the highest distractor pull rate (82.0%), highlighting a tradeoff between retrieval precision and robustness.

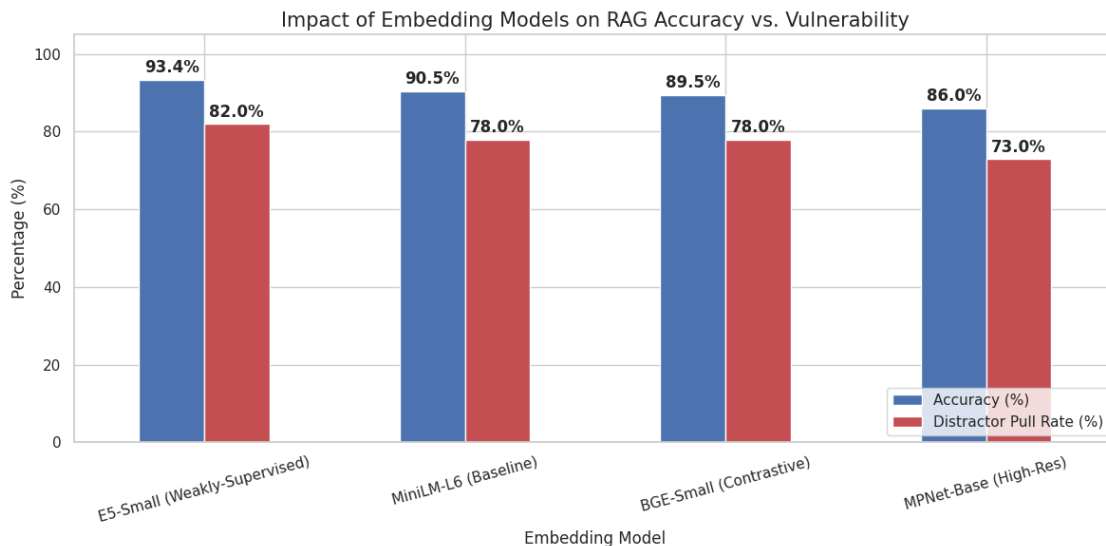


Figure 3: Impact of embedding models on accuracy and distractor pull rate. Higher-performing embeddings achieve greater accuracy but are more susceptible to distractor retrieval, revealing an accuracy–robustness tradeoff.

4.4 Effect of Chunking Strategy

Chunk size has a non-linear impact on performance. As shown in Figure 4, accuracy follows an inverted U-shaped curve, while distractor pull rate decreases as chunk size increases.

- Small chunks (128 tokens) achieve 83.0% accuracy, likely due to insufficient contextual information.
- Medium chunks (512 tokens) achieve optimal performance at 90.7%.

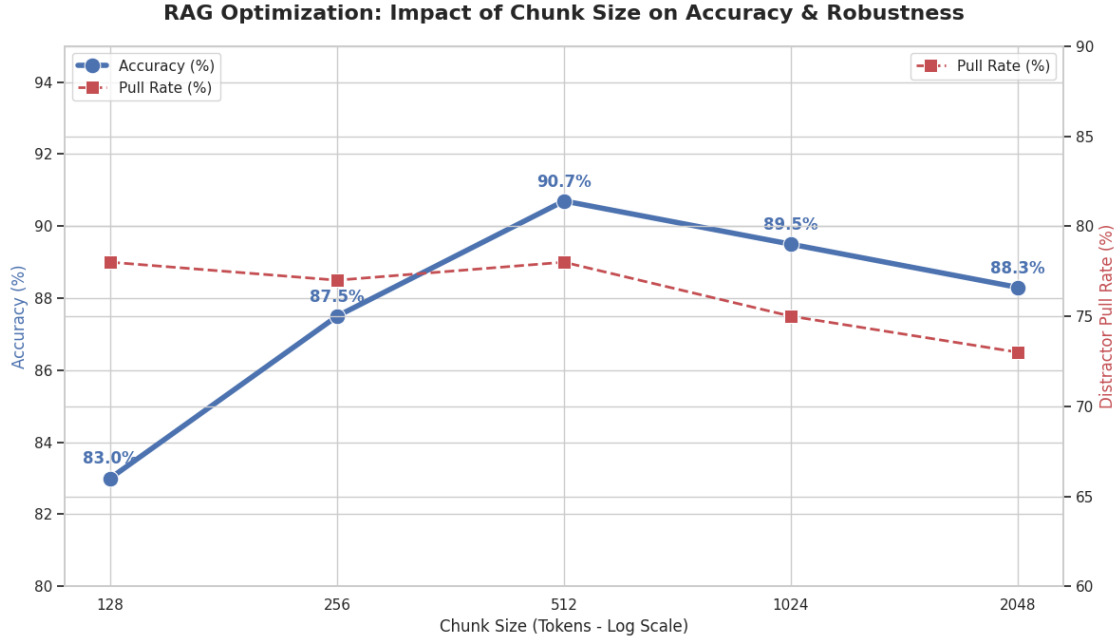


Figure 4: Impact of chunk size on RAG performance. Accuracy peaks at 512 tokens, while distractor pull rate decreases for larger chunks, revealing a tradeoff between contextual completeness and robustness.

- Large chunks (2048 tokens) degrade to 88.3%, suggesting context dilution and increased noise.

This result demonstrates that both insufficient and excessive context negatively impact retrieval effectiveness.

4.5 Model Scaling Effects

To evaluate robustness across model sizes, we compare Mistral-Large and Mistral-Small under both RAG and no-RAG conditions. While Mistral-Large achieves higher overall accuracy (90.7% vs. 84.0% under FAISS), both models benefit from retrieval augmentation.

RAG improves performance by approximately 11.8 percentage points for the large model and 6.0 percentage points for the smaller model, indicating that retrieval augmentation provides consistent benefits across model scales, though larger models are better able to leverage retrieved context.

4.6 Key Observations

Across all experiments, several consistent patterns emerge:

- RAG substantially reduces hallucination compared to no-retrieval baselines.
- Retrieval quality is the dominant factor influencing performance.
- Embedding models introduce a tradeoff between accuracy and robustness to distractors.
- Chunk size must balance contextual completeness and noise.
- High distractor pull rates (70–82%) do not necessarily result in incorrect answers, indicating that LLMs can partially filter noisy context.

5 Discussion

The results of our experiments provide several important insights into the behavior of Retrieval-Augmented Generation (RAG) systems in small, controlled environments. In particular, they highlight the interplay between retrieval quality, model reasoning, and robustness to adversarial noise.

5.1 Embedding Performance is Dataset-Dependent

One surprising finding is that the state-of-the-art embedding model BGE-Small underperforms the simpler MiniLM-L6 baseline (89.5% vs. 90.5% accuracy). This contradicts the common assumption that embeddings optimized on large-scale benchmarks will consistently outperform lighter models in all settings.

This result suggests that embedding effectiveness is highly dependent on dataset characteristics. In small, controlled corpora, simpler models may generalize better due to reduced overfitting to large-scale semantic patterns. Consequently, embedding selection should be empirically validated rather than assumed based on benchmark performance.

5.2 Necessity of Retrieval

Although retrieval quality is a dominant factor in overall performance, it does not fully eliminate hallucination. We observe multiple cases where the correct document is successfully retrieved ($\text{Recall@5} \approx 100\%$), yet the model still produces incorrect answers.

This phenomenon, which we term *hallucination despite correct context*, highlights a critical limitation of RAG systems: retrieval and reasoning are distinct processes. Even when provided with accurate context, the LLM may fail to correctly interpret or utilize the information. This suggests that improving retrieval alone is insufficient, and future work must also address reasoning robustness within LLMs.

5.3 Accuracy-Robustness Tradeoff

A key contribution of this work is the identification of a tradeoff between accuracy and robustness to distractors. The highest-performing configuration (E5-Small, 93.4% accuracy) also exhibits the highest distractor pull rate (82.0%), indicating increased susceptibility to misleading context.

Conversely, more robust configurations, such as hybrid retrieval, achieve lower distractor pull rates (71.0%) at a slight cost to accuracy. This tradeoff suggests that optimizing for maximum accuracy may inadvertently increase vulnerability to adversarial noise.

In practical deployments, this implies that system designers must balance peak performance with robustness requirements, depending on the reliability constraints of the application domain.

5.4 Effectiveness of Hybrid Retrieval

Hybrid retrieval methods provide a compelling balance between performance and robustness. By combining semantic (FAISS) and lexical (TF-IDF) signals, hybrid systems reduce the likelihood of retrieving distractor documents while maintaining high accuracy.

Our results show that hybrid retrieval achieves competitive accuracy (90.0%) while significantly reducing distractor pull rate. This makes it a strong candidate for real-world deployments where document quality cannot be guaranteed.

5.5 Implications for Low-Resource Systems

Our findings also demonstrate that effective RAG systems can be built without heavy computational resources. BM25-based retrieval achieves high recall and competitive accuracy (87.7%) without requiring GPU acceleration or vector indexing.

This highlights the viability of lightweight RAG pipelines for edge deployments and small organizations, where computational constraints limit the use of large-scale vector databases.

5.6 Summary of Insights

Overall, our analysis reveals that RAG performance in small-scale settings is governed by three interacting factors:

- Retrieval quality determines the availability of relevant context.
- Model reasoning determines how effectively that context is used.
- Dataset structure influences robustness to adversarial noise.

Understanding and balancing these factors is essential for designing reliable RAG systems in constrained environments.

6 Error Analysis

To better understand failure modes in retrieval-augmented systems, we analyze representative errors observed during evaluation. These examples highlight how distractor documents influence model behavior, even when retrieval and reasoning mechanisms are partially effective.

6.1 Factual Overriding (Grounded Hallucination)

Question: How deep within the Earth do mantle plumes originate?

Retrieved Context: A distractor document modified the correct depth (3,000 km) to an incorrect value (5,000 km).

Model Output: The model confidently answered “5,000 km,” explicitly citing the retrieved context.

Ground Truth: 3,000 km

Analysis: This example illustrates a *grounded hallucination*, where the model produces an incorrect answer that is nonetheless supported by retrieved evidence. Because the system is designed to prioritize external context over internal knowledge, incorrect retrieval directly propagates into the final answer. This demonstrates that RAG systems are only as reliable as their retrieval component.

6.2 Noise-Induced Confusion

Question: By what percentage has atmospheric CO₂ increased since the pre-industrial era?

Retrieved Context: Both correct (50%) and distractor (80%) values were present in the retrieved documents.

Model Output: The model identified 50% as the correct value but included a hedge referencing both values.

Ground Truth: 50%

Analysis: Although the model successfully identified the correct answer, the presence of conflicting information introduced uncertainty into the response. The model attempted to reconcile the inconsistency by explicitly acknowledging both values. This highlights how distractor documents can reduce answer clarity and confidence, even when the correct information is retrieved.

6.3 Attempted Error Correction

Question: Which law describes magnetism alongside Coulomb’s law?

Retrieved Context: A distractor document incorrectly substituted Ampère’s force law with “Faraday’s force law.”

Model Output: The model correctly identified Ampère’s law and explicitly noted that the retrieved context appeared incorrect.

Ground Truth: Ampère’s force law

Analysis: In this case, the model attempted to override incorrect retrieved information using its internal knowledge. However, the presence of conflicting context increased response complexity, leading to a failure when scoring the response. This example demonstrates that RAG introduces a cognitive burden: the model must not only generate answers but also resolve inconsistencies within retrieved context.

6.4 Summary of Failure Modes

Across these examples, we identify three recurring error types:

- **Grounded Hallucination:** Incorrect answers supported by misleading retrieved context.
- **Contextual Ambiguity:** Conflicting information leading to hedged or unclear responses.
- **Reasoning Conflict:** Tension between retrieved context and internal model knowledge.

These findings reinforce that while RAG improves factual grounding, it also introduces new failure modes that arise from the interaction between retrieval quality and model reasoning.

7 Limitations

While our study provides a controlled and systematic analysis of RAG behavior in small-scale settings, several limitations should be considered when interpreting the results.

7.1 Dataset Scale and Scope

Our experimental corpus is intentionally limited to 100 documents to enable fine-grained control over variables such as retrieval noise and chunking strategies. While this micro-scale design allows for precise analysis, it may not fully capture the complexity of large-scale production systems that operate over millions of documents.

Additionally, the dataset covers a relatively narrow set of topics, which may not reflect long-tail knowledge distributions or highly heterogeneous domains where retrieval becomes more challenging.

7.2 Synthetic Distractor Design

The distractor documents used in this study are synthetically generated to introduce controlled adversarial noise, such as entity swaps and numerical perturbations. While effective for isolating specific failure modes, these distractors may not fully represent the diversity and subtlety of real-world misinformation.

In particular, real-world noise often involves nuanced semantic shifts, stylistic variations, and incomplete or ambiguous statements, which are not fully captured by templated synthetic modifications.

The judge model may exhibit systematic preferences or reasoning errors, potentially affecting the reliability of the assigned scores. Furthermore, although the scoring framework supports continuous values, the implemented evaluation primarily reflects binary outcomes (correct vs. incorrect), which may not fully capture partial correctness or nuanced reasoning present in model outputs.

7.3 Generalization to Other Models

Most experiments in this study are conducted using a single primary model (Mistral-Large), with limited evaluation on smaller models. While results indicate that RAG benefits transfer across model sizes, further experimentation with a broader range of architectures would be necessary to fully generalize these findings.

7.4 Summary

These limitations highlight the tradeoff between experimental control and real-world complexity. While our controlled setup enables detailed analysis of RAG behavior, future work is needed to validate these findings in larger, more diverse, and less structured environments.

8 Conclusion

In this work, we presented a controlled, systematic study of Retrieval-Augmented Generation (RAG) in small-scale, highly curated datasets. Our results demonstrate that RAG significantly reduces hallucination, improving accuracy by up to 11.8 percentage points over no-retrieval baselines. However, these gains are highly sensitive to retrieval configuration choices.

We find that RAG effectiveness depends critically on factors such as embedding selection and chunking strategy. In particular, chunk size exhibits a non-linear relationship with performance, with intermediate sizes (512 tokens) providing the best balance between contextual completeness and noise. Furthermore, embedding models introduce a fundamental tradeoff between accuracy and robustness: while E5-Small achieves the highest accuracy (93.4%), it is also the most susceptible to distractor contamination, whereas more robust models achieve lower peak performance.

Our findings highlight that the dominant driver of hallucination is not simply the presence of retrieval, but the quality and behavior of the retrieval system itself. Notably, state-of-the-art embedding models do not consistently outperform simpler baselines in small-scale settings, emphasizing the need for empirical validation rather than reliance on benchmark performance.

From a practical perspective, these results suggest that developers should carefully tune and evaluate retrieval configurations on their target corpus, rather than assuming that standard RAG pipelines or top-performing models will generalize effectively. Overall, this work provides a reproducible framework and actionable insights for designing reliable RAG systems in constrained, real-world environments.

A Code Availability

The full implementation, dataset construction pipeline, and experimental framework are publicly available at: <https://github.com/NabarunKar/rag-hallucination-benchmark>.